Roll No.....

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Program: M.Tech.(CSE)
Name of the Course: Intrusion Detection System

Semester: 3rd
Course Code: MCS 324

Time: 03 Hours
Note:-

Maximum Marks: 100

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1		
a)	Define Network Attacks which are used to compromise the three important triad. Discuss how these triads are important to model designed to guide policies for information security within an organization? (20 marks)	CO1
b)	What do you understand about Network Probes? List the name and briefly describe publically available tools that are used for network probing.	
c)	What is Payload attack? Classify and define the categories of Payload attack.	
Q2		
a)	Discuss pros and cons of supervised and unsupervised anomaly detection approach. (20 marks)	CO2
b)	How the network-based Intrusion detection system is useful to collect and analyse the data directly from the Network? Discuss about the SNMP.	
c)	How System Call (SC) sequences have proved to be an effective information source in host-based intrusion detection? Even though system-calls prove to be a good information source for intrusion detection at the application layer, they still face several limitations, write about those limitations.	
Q3		
a)	Discuss about the labelling. What are the important operating modes are defined for anomaly detection techniques? (20 marks)	CO3
b)	There are types of training sets, anomaly detection methods can be categorized based on the method they use to report the anomalies. Discuss about the Type of output to report the anomalies.	
c)	Differentiate centralized IDS architecture with Hierarchical IDS architecture.	
Q4		
a)	Discuss about fuzzy membership function and demonstrate it by mathematical notation with suitable example. (20 marks)	CO4
b)	What are the reasons distributed IDS superseded by cooperative intrusion detection?	
c)	How Profile-Based Matching Works? Discuss about some major advantages and downsides of profile-based intrusion detection.	
Q5		
a)	Are mobile agents suitable in intrusion detection? If yes, then Name the mobile agent based IDS. (20 marks)	CO5
b)	What are the reasons distributed IDS superseded by cooperative intrusion detection?	
c)	Discuss about policies, standards, guidelines, procedures and baselines for Intrusion detection and prevention program.	

20/12/24
END
EVENING